

On Valuable Second-Order Mutants for Mutation Testing

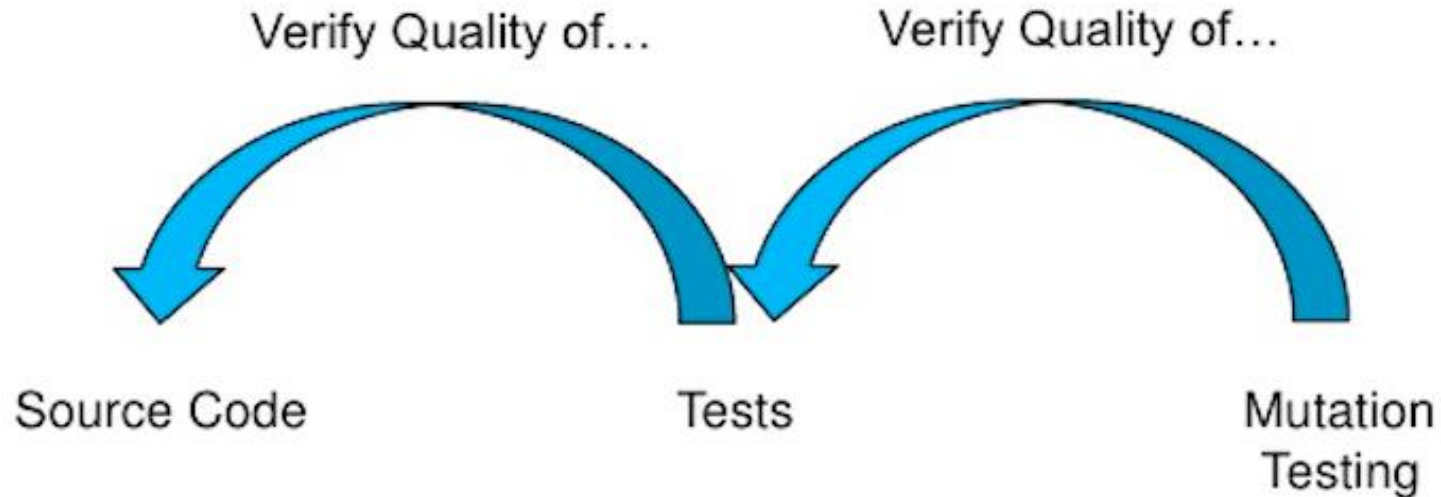
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LabSoft Seminar. Mai 28th, 2026

Outline

- ❑ Mutation testing
- ❑ Higher-order mutants
- ❑ Mutants' generation
- ❑ Exhaustive search for valuable second-order mutants
- ❑ Valuable second-order mutants' characterization
- ❑ Threats to Validity
- ❑ Related work
- ❑ Conclusion

Introduction



Example of a mutant

Original:

```
public class Taxes {  
  
    double simpleTax(double amount) {  
  
        return amount * 0.2;  
    }  
}
```

Example of a mutant

Mutation place:

```
public class Taxes {  
  
    double simpleTax(double amount) {  
  
        return amount * 0.2;  
    }  
}
```

Example of a mutant

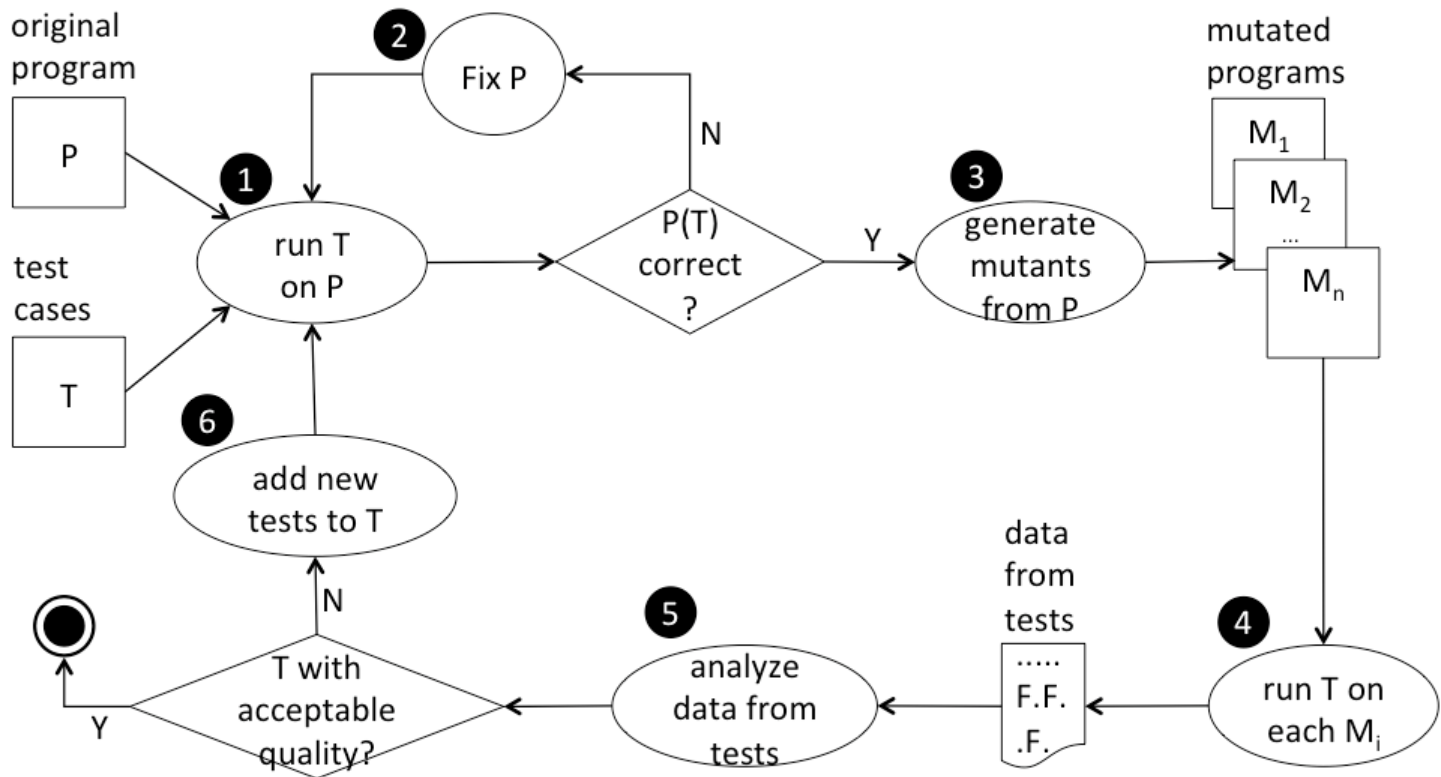
* → +

```
public class Taxes {  
  
    double simpleTax(double amount) {  
  
        return amount + 0.2;  
    }  
}
```

Mutation Testing

- ❑ Introducing **artificial changes** into original source code
 - Syntactic changes (**mutations**)
 - Representing real common programming mistakes
 - Changed programs are called **mutants**
- ❑ Running test cases on mutants
 - Result different from original: mutant **killed**
 - Otherwise: **alive**

Mutation testing process



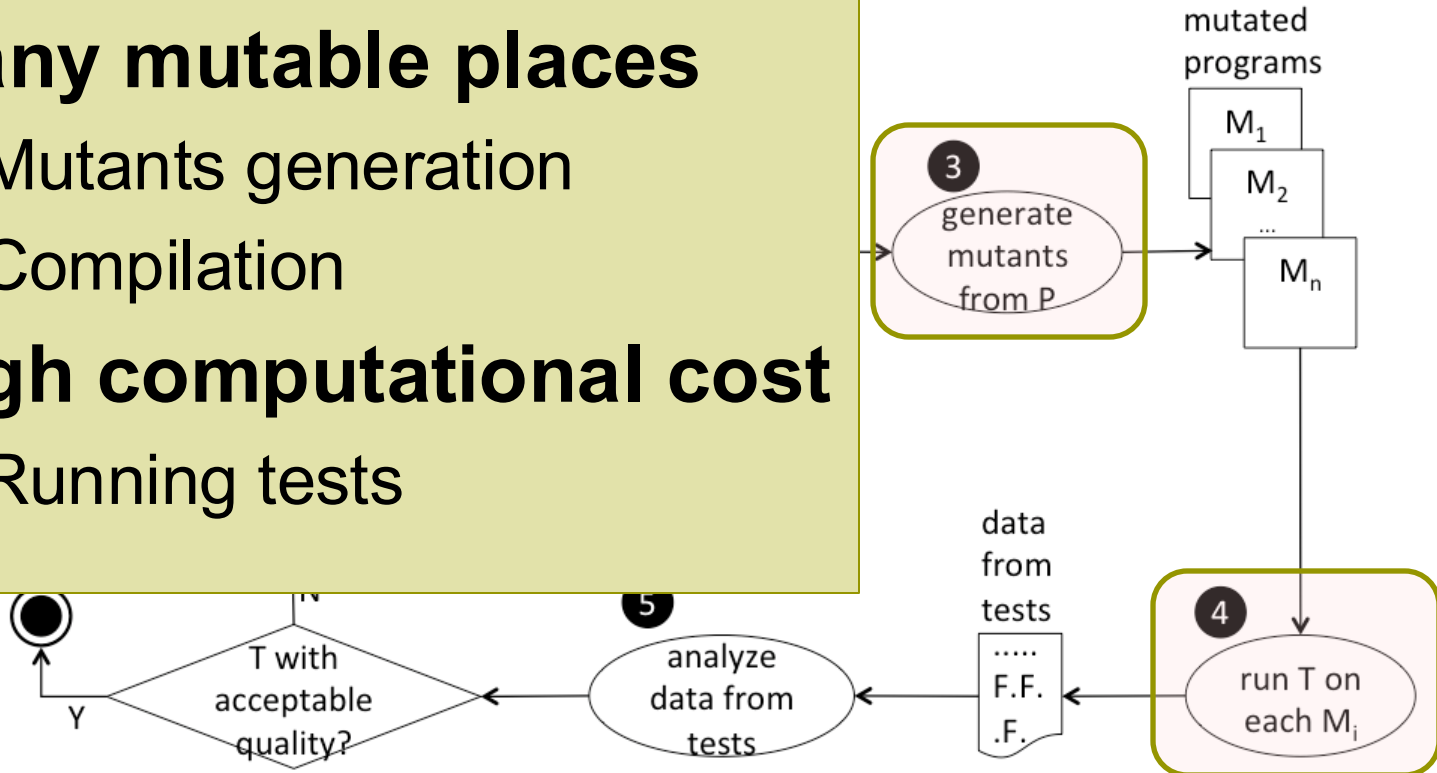
Mutation testing drawbacks

3. Many mutable places

- Mutants generation
- Compilation

4. High computational cost

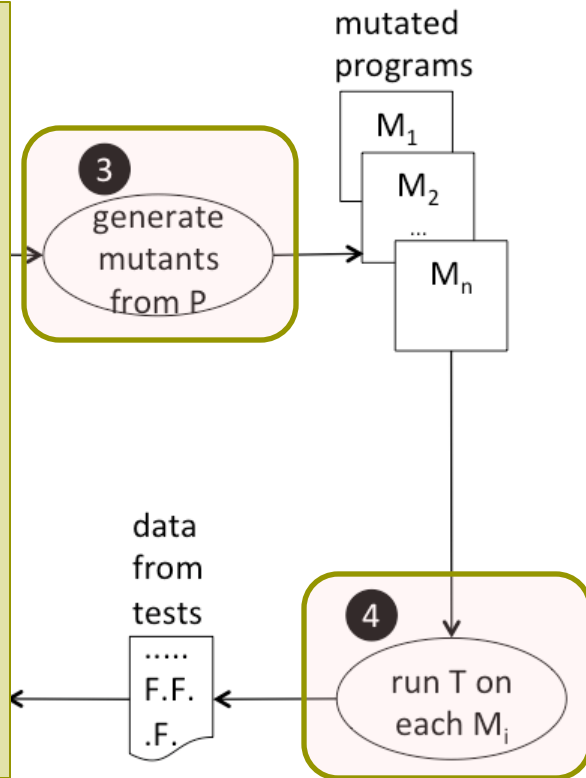
- Running tests



Mutation testing drawbacks

❑ Cost reduction techniques

- Number of mutants
- Number of test cases
- Test case prioritization
- **Higher-order mutation testing**



Higher-order Mutants

Higher-order mutants

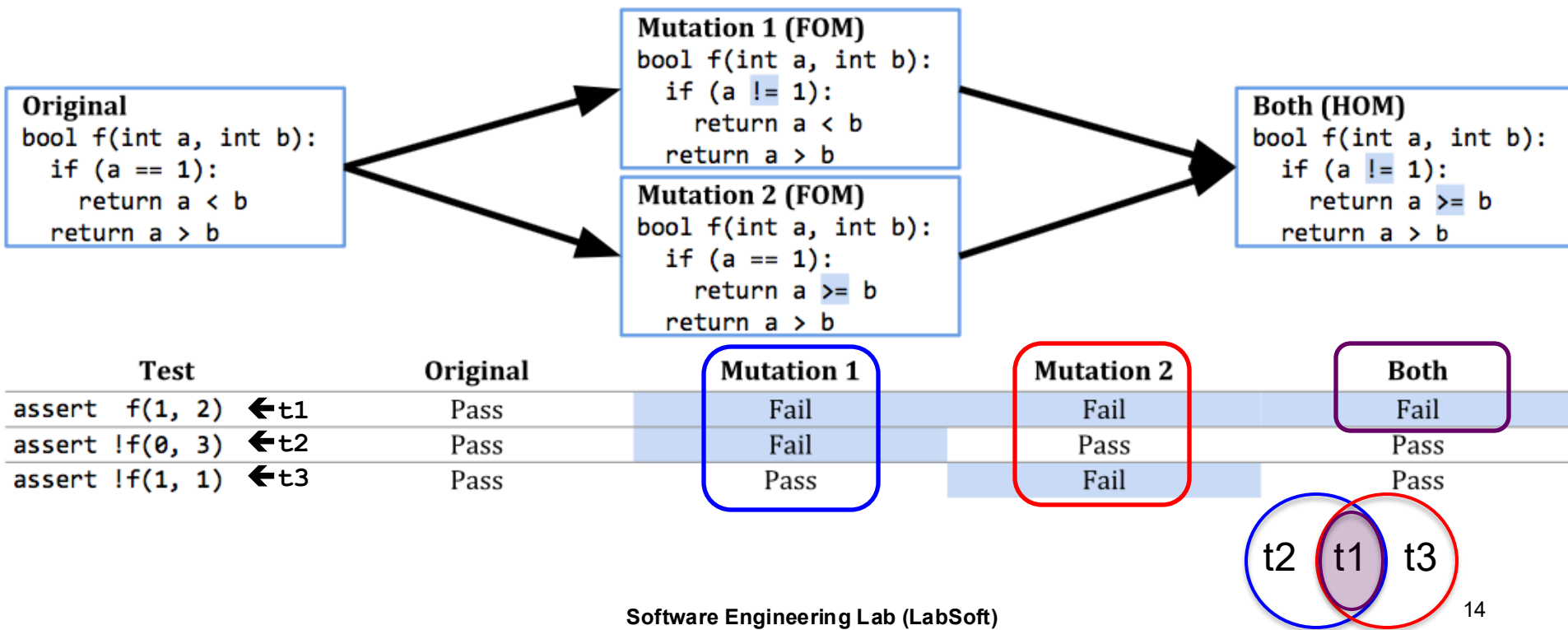
- 1 mutation → first-order mutant (**FOM**)
- > 1 mutations → higher-order mutant (**HOM**)
 - 2 mutations → 2nd order mutant (**2OM**)

- **n** mutations can generate
 - $\sim C_2^n$ 2OMs
 - $\sim C_3^n$ 3OMs
 - $\sim 2^n$ HOMs (exponential)

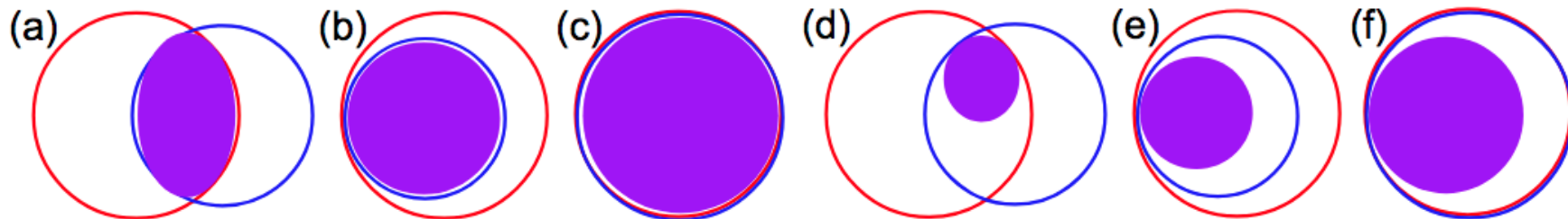
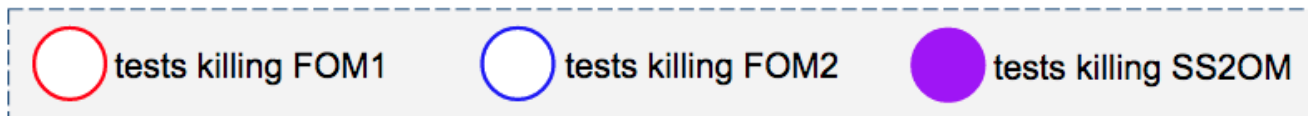
Higher-order mutants

- Most are **more likely to be killed** than FOMs
- But few are **harder** to be killed
 - are **more representative**
 - represent **subtler faults**
 - mutations can (partially) **mask** each other

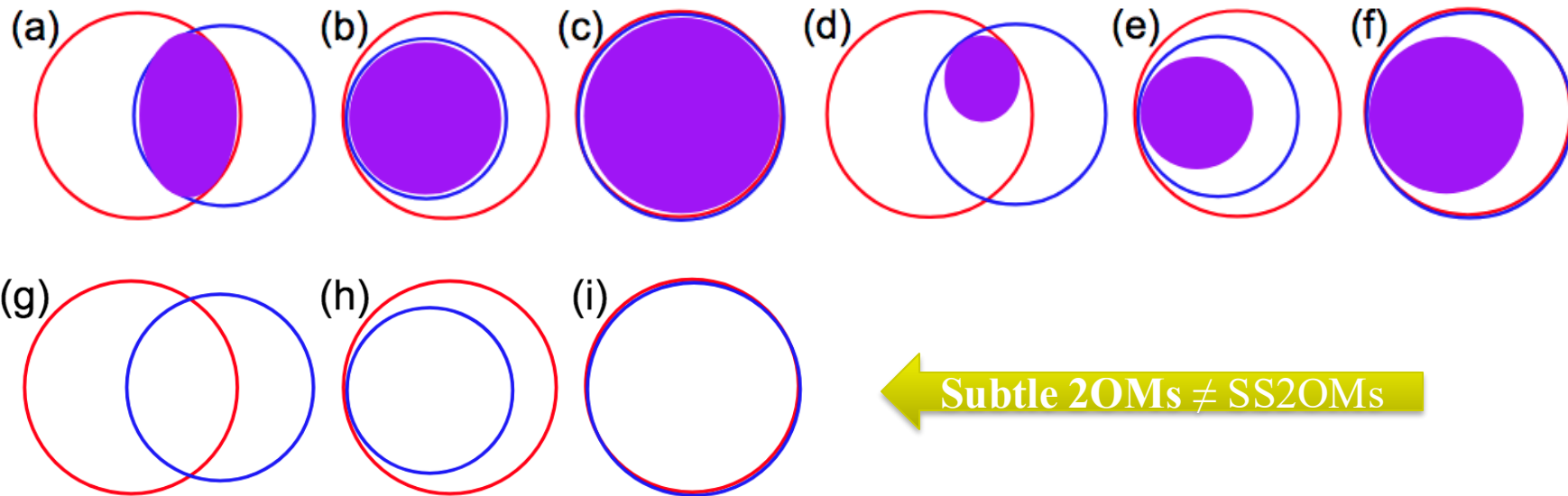
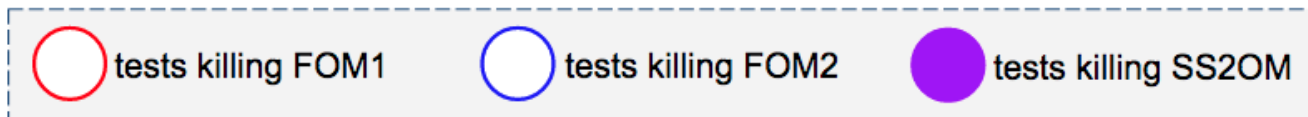
A 2OM subsumption example



Venn Diagram “shapes”



Venn Diagram “shapes”



Subtle 2OMs $\neq$ SS2OMs

SSHOMs research

- ❑ Exponential search space
- ❑ Dynamic search (running test suite)
- ❑ Trends:
 - Find as many SSHOMs as possible
 - Evolutionary algorithms
 - ❑ Genetic Algorithms
 - ❑ Hill Climbing
 - Multi-objective search

Research Method

Main goal

- Understand which **characteristics** of SS2OMs
- Calculate mutants **reduction** via SS2OMs

Dataset - Java subject systems

System	Version	LOC	# Tests	JUnit
Vending Machine	Exceptions	~100	35	4
Triangle	n/a	34	12	4
Monopoly	n/a	1,181	124	3
Commons CSV	1.8	~2k	325	4
Commons CLI	1.4	2,699	318	4
ECal	2003.10	3,626	224	3
Commons Validator	1.6	7,409	536	4
Gson	2.9.0	> 10k	1,089	3 and 4
Chess	n/a	4,924	930	3 and 4
Joda-Time	2.12.7	~30k	4,232	4

Mutation operators implemented

Operator	Name	Examples	
		Original	Mutants
AOR	Arithmetic Operator Replacement	<code>a + b</code>	<code>a - b</code> <code>a * b</code> <code>a / b</code> <code>a % b</code>
ROR	Relational Operator Replacement	<code>a < b</code>	<code>a > b</code> <code>a <= b</code> <code>a >= b</code> <code>a == b</code> <code>a != b</code>
LCR	Logical Connector Replacement	<code>a b</code>	<code>a && b</code>
SBR	Statement Block Removal	<code>a = b+1;</code> <code>if (...) {</code> <code>...</code> <code>}</code>	<code>∅</code> <code>∅</code>

FOMs generated

System	AOR	LCR	ROR	SBR	Total
Vending Machine	4	1	40	12	57
Triangle	36	7	85	10	138
Monopoly	120	8	380	358	866
Commons CSV	208	56	250	411	925
Commons CLI	108	80	355	539	1,082
ECal	136	9	315	753	1,207
Commons Validator	324	174	1,675	1,024	3,197
Gson	536	211	1,430	1,535	3,712
Chess	1,876	244	1,890	1,309	5,287
Joda-Time	4,300	437	6,985	4,701	16,333

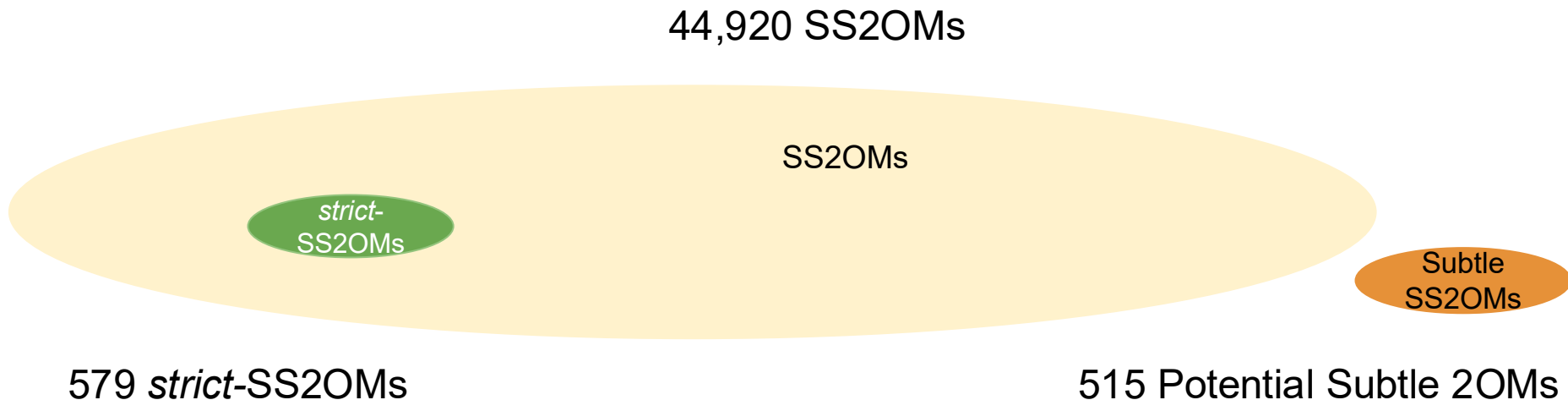


Results

Valuable 2OMs found

System	FOMs	2OMs	SS2OMs	<i>strict-</i>	Pot. Subtle
Vending Machine	57	1,488	25	-	-
Triangle	138	9,086	393	5	13
Monopoly	866	372,885	3,324	58	46
Commons CSV	925	425,531	4,430	15	11
Commons CLI	1,082	582,685	1,852	41	8
ECal	1,207	727,556	1,421	26	-
Commons Validator	3,197	5,102,330	17,546	121	54
Gson	3,712	6,880,416	6,970	140	79
Chess	5,287	13,952,357	8,959	173	304

Valuable 2OMs found



Research questions

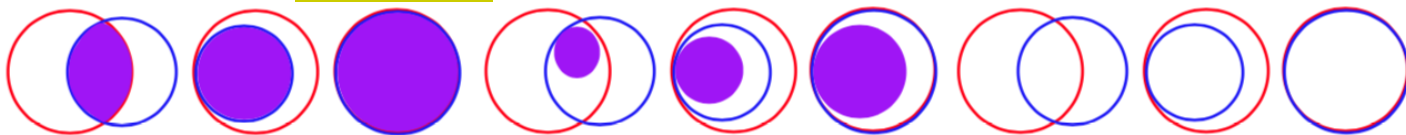
- ❑ **RQ2:** ❑ is there a pattern for the **distribution** of the mutations in the analyzed Valuable 2OMs?
- ❑ **RQ3:** What are the **most common killing tests** Venn Diagram “shapes” of the Valuable 2OMs found?
- ❑ **RQ6:** What is the **reduction in the number of mutants** the SS2OMs achieve by replacing their constituent FOMs?

RQ2: Mutations distribution

System	SS2OMs			strict-SS2OMs			Pot. Subtle 2OMs		
	SM	SC	DC	SM	SC	DC	SM	SC	DC
Vending Machine	20	5	-	-	-	-	-	-	-
Triangle	393	-	-	5	-	-	13	-	-
Monopoly	1,110	1,470	596	13	3	42	30	-	16
Commons CSV	3,422	736	272	13	2	-	11	-	-
Commons CLI	1,627	199	26	41	-	-	8	-	-
ECal	761	538	122	22	4	-	-	-	-
Commons Validator	6,416	10,442	713	106	15	-	54	-	-
Gson	4,291	2,541	138	139	1	-	48	31	-
Chess	8,738	201	38	165	8	-	304	-	-

86.42%

RQ3: Venn



Vending Machine	-	7	18	-	-	-	-	-	-
Triangle	38	143	207	-	4	1	-	-	13
Monopoli	1	414	2,851	1	2	55	-	-	46
Commons CSV	-	116	4,299	-	1	14	3	-	8
Commons CLI	9	429	1,373	-	30	11	5	-	3
ECal	8	223	1,164	-	11	15	-	-	-
Commons Validator	32	1,526	15,867	-	20	101	18	-	36
Gson	60	1,079	5,691	-	73	67	25	-	54
Chess	265	1,674	6,847	5	83	85	172	-	132

Mutants reduction - example

FOMs

m1

m2

m3

m4

m5

m6

m7

m8

Mutants reduction - example

FOMs

SS2OMs

m1

[m1,m2]

m2

[m3,m4]

m3

[m5,m6]

m4

m5

m6

m7

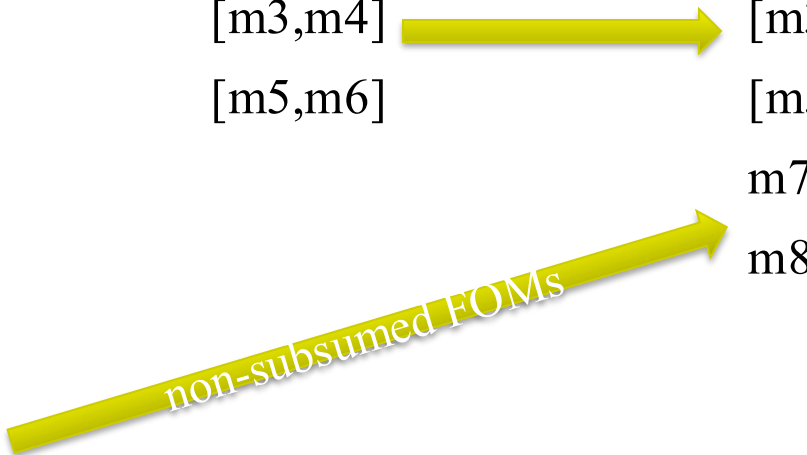
m8

Mutants reduction - example

FOMs	SS2OMs	Resulting mutants
m1	[m1,m2]	[m1,m2]
m2	[m3,m4]	[m3,m4]
m3	[m5,m6]	[m5,m6]
m4		
m5		
m6		
m7		
m8		

Mutants reduction - example

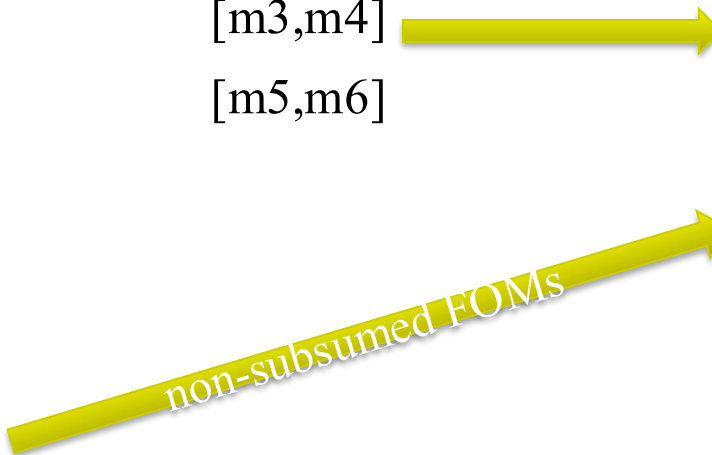
FOMs	SS2OMs	Resulting mutants
m1	[m1,m2]	[m1,m2]
m2	[m3,m4]	[m3,m4]
m3	[m5,m6]	[m5,m6]
m4		m7
m5		m8
m6		
m7		
m8		



non-subsumed FOMs

Mutants reduction - example

FOMs	SS2OMs	Resulting mutants	Reduction
m1	[m1,m2]	[m1,m2]	3 out of 8 mutants (37.5%)
m2	[m3,m4]	[m3,m4]	
m3	[m5,m6]	[m5,m6]	
m4		m7	
m5		m8	
m6			
m7			
m8			



non-subsumed FOMs

With much more SS2OMs

FOMs

SS2OMs

Resulting mutants

m1 [m1,m2], [m1,m4], ...,

m2 [m2,m3], ..., [m2,m8],

m3 [m3,m4],..., [m3,m8],

m4 [m4,m5], ..., [m4,m8],

m5 [m5,m6], [m5,m7], ...,

m6 [m6,m7], [m6,m8],

m7 [m7,m8], [m1,m7], ...,

m8 [m2,m4], ..., [m5,m8]

??????????

**Minimum
Set Cover
Problem**

MSC by Integer Linear Programming

$$\begin{array}{ll}\text{minimize} & \sum_{j=1}^n x_j \\ \text{subject to} & \sum_{j:v \in S_j} x_j \geq 1 \quad \forall v \in U, \\ & x_j \in \{0, 1\} \quad \forall j \in \{1, \dots, n\}.\end{array}$$

RQ6: Mutants reduction (1/2)

System	SS2OMs	SS2OMs red.	SS2OMs discarded	Non-sub. FOMs	Resulting mutants
Vending Machine	25	17	32.00%	32	49
Triangle	393	52	86.77%	36	88
Monopoli	3,176	270	91.50%	387	657
Commons CSV	4,430	234	94.72%	503	737
Commons CLI	1,066	311	70.83%	545	856
ECal	1,488	366	75.40%	568	934
Commons Validator	17,571	944	94.63%	1,464	2,408
Gson	6,970	976	86.00%	1,978	2,954
Chess	8,977	1,402	84.38%	2,832	4,234
Overall	44,920	4,675	89.61%	8,253	12,928

RQ6: Mutants reduction (2/2)

System	FOMs	Resulting mutants	Mutants reduction
Vending Machine	57	49	14.04%
Triangle	138	88	36.23%
Monopoli	866	657	24.13%
Commons CSV	925	737	20.32%
Commons CLI	1,082	856	31.05%
ECal	1,213	934	22.54%
Commons Validator	3,197	2,408	24.52%
Gson	3,712	2,954	20.42%
Chess	5,319	4,234	20.38%
Overall	16,471	12,928	22.37%

Related work

2nd order mutation testing

- 2OMs sampling
 - Random
 - LastToFirst
 - Different operators
 - etc.
- [Polo et al., 2009], [Kintis et al., 2010], [Madeisky et al., 2013] ...

SSHOMs

- [Jia et al., 2008 and 2009]
 - First search-based algorithms proposed
 - SSHOMs examples
- [Harman et al., 2014]
 - Reduction via greedy minimization
- [Wong et al., 2020]
 - Search for all possible SSHOMs via Variational Execution
 - Search_{pri} using some characteristics

Conclusion



Questions?

